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In : **Software Engineering**

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CHAPTER 1
INTRODUCTION

1.1 Purpose

This document provides a comprehensive overview of the platform designed to facilitate interaction and management within an educational institution. It outlines the system's objectives, functionalities, and requirements to guide development and implementation.

1.2 Product scope

The platform will connect students, teachers, and administration. It provides students access to their grades, course information, and updates. Additionally, it offers a chatbot for quick answers to common queries and a functionality for students to appeal their grades. Teachers can manage marks, deliberate on appeals, and the system will automatically calculate final grades based on predefined criteria. .

1.3 Glossery

- **SRS (Software Requirements Specification)**: A document detailing the functional and non-functional requirements of a software system.

- **DB (Database)**: A collection of electronically organized data used by the system to manage user information, courses, and grades.

- **CRUD (Create, Read, Update, Delete)**: The basic operations used to manage data in a database.

- **API (Application Programming Interface)**: A set of functions allowing different applications to communicate with each other.

- **User Management Module**: The component of the system that handles authentication and profiles for students, teachers.

- **Course and Grade Management**: A feature allowing teachers to manage courses and enter student grades, while students can view their courses and results.

1.4 References

-Educational institution policies and grading rules.
-Technical references for technologies and frameworks used (e.g., databases, APIs).

1.5 Overview

This SRS will detail the functionalities, system architecture, external interfaces, and both functional and non-functional requirements of the platform.

CHAPTER 2

GENERAL DESCRIPTION

2.1 Product perspective

The platform will be a web-based application integrating several services to create an efficient academic management system. The key components include:

2.1.1 User Management Module

Handles student, teacher, and administration profiles and authentication using digital ID cards.

2.1.2 Course and Grade Management

Teachers input grades, students view their results and course information, and the platform allows for grade appeals and updates.

2.2 Product function

2.2.1 User Authentication and Authorization

Secure login for students, teachers, and administration, supporting the use of digital ID cards.

2.2.2 Course Management

Teachers can create, update, and manage courses and course-related materials.

2.2.3 Grade Management

Teachers can input grades, the system automatically calculates final grades, and students can appeal grades.

2.2.4 Reporting and Analytics

Provides reports on student performance and other relevant metrics for teachers and administrators.

2.3 User Characteristics

2.3.1 Students

Users who access their grades, view courses, appeal grades, and interact with the chatbot using their digital ID cards.

2.3.2 Teachers

Users who input student marks, manage course details, deliberate on appeals, and receive assistance from the chatbot.

2.4 Constraints

-Data privacy and security regulations (e.g., GDPR compliance).
-System must support high availability and load balancing to handle multiple users simultaneously.

2.5 Assumptions and Dependencies

-The institution provides the necessary grading rules and policies.
-Availability of a stable internet connection for accessing the platform.
-The system depends on external APIs for chatbot functionality and integration with other services if necessary.

CHAPTER 3

SPECIFIC REQUIREMENTS

3.1 Functional Requirements

3.1.1 User Authentication

-Users must be able to register and log in securely using their digital ID cards.
-Role-based access controls will differentiate students, teachers, and administrators.

3.1.2 Course Management

-Teachers can create, update, and delete courses.
-Students can view their enrolled courses.

3.1.3 Homework implementation

- Our application has a homeWorks management that corrects homeworks ,

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

-The platform must handle at least 1,000 concurrent users.

3.2.2 Security Requirements

-Data encryption for all sensitive information (e.g., grades, user credentials).
-Regular security audits to ensure data protection and system integrity.

3.2.3 Usability Requirements

-The platform should have a user-friendly interface accessible smartphones).
-Accessibility compliance to accommodate users with disabilities.

3.2.4 Reliability Requirements

The platform should maintain 99.9A backup system should ensure data recovery in case of failure.

CHAPTER 4

EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

A dashboard for students, teachers, and administrators with role-specific features, including digital ID card support.

4.2 Hardware Interfaces

The platform will be accessible via standard web browsers on any compatible device.

4.3 Software Interfaces

-Integration with databases for data storage (e.g., MySQL, PostgreSQL).
-API integration for the chatbot service and ID card authentication.

4.4 Communication Interfaces

The platform will use HTTPS protocol for secure communication.

CHAPTER 5

OTHER REQUIREMENTS

5.1 Legal and Regulatory Requirements

The platform must comply with data privacy regulations (e.g., GDPR) to protect user information.

5.2 Documentation Requirements

User manuals for students, teachers, and administrators to guide platform usage.

CHAPTER 6

TOOLS

6.1 Frontend

we used Flutter for making this application for: - Cross-Platform Development - Fast Development - Open Source

6.2 Backend

We used Node.js, MongoDB for our database, JWT, and testing end points using Postman.